

Math Camp

Lecture 3

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Coordinates

- Let $\mathbf{V}$ be a real vector space of dimension n , and let $A = \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ be a basis for $\mathbf{V}$.
 - $\mathbf{v} \in \mathbf{V} \Rightarrow \exists! \{\alpha_k^v\}_{k=1}^n$ s.t. $\mathbf{v} = \sum_{k=1}^n \alpha_k^v \mathbf{a}_k$
 - The scalars $\{\alpha_1^v, \dots, \alpha_n^v\}$ comprise the coordinate representation of $\mathbf{v}$ with respect to the basis A .

Kronecker delta

- Given two sets X and Y , $\delta : X \times Y \rightarrow \{0, 1\}$ is defined by $\delta_{xy} = 1$ iff $x = y$.
 - $X = Y = \mathbb{Z}$ is illustrative: $\delta_{ij} = 1 \Leftrightarrow i = j$

- Consider the n -dimensional vector space $\mathbb{R}^n$
- $\mathbf{x} \in \mathbb{R}^n$ implies $\mathbf{x} = (x_1, \dots, x_n)$ with $x_i \in \mathbb{R}$
 - Think of $\mathbf{x}$ as a column vector

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

- The *canonical basis* for $\mathbb{R}^n$ is $E = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$, where $\mathbf{e}_j = (e_{1j}, \dots, e_{nj})$ and $e_{ij} = \delta_{ij}$

- Let $\mathbf{V}$ be a real vector space of dimension n , and let $A = \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ be a basis for $\mathbf{V}$.
- Given $\mathbf{v} \in \mathbf{V}$, its coordinate representation against A defines an n -tuple $\alpha^v \in \mathbb{R}^n$.
- Define $\varphi_A : \mathbf{V} \rightarrow \mathbb{R}^n$ by $\varphi_A(\mathbf{a}_i) = \mathbf{e}_i$ and extend linearly.
 - φ_A is an isomorphism, and $\varphi_A(\mathbf{v}) = \alpha^v$.
 - $\mathbf{V} \simeq \mathbb{R}^n$
- φ_A depends on the choice of A .

- A real $m \times n$ matrix is a rectangular array of real numbers with m rows and n columns.
 - An n -dimensional vector is an $n \times 1$ matrix
- The collection of real $m \times n$ matrices may be written as $\mathbb{R}^{m \times n}$
- If $M \in \mathbb{R}^{m \times n}$ then $M = (m_{ij})$, where $i = 1, \dots, m$ and $j = 1, \dots, n$,

$$M = \begin{pmatrix} m_{11} & m_{12} & \dots & m_{1n} \\ m_{21} & m_{22} & \dots & m_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ m_{m1} & m_{m2} & \dots & m_{mn} \end{pmatrix}$$

- $\mathbb{R}^{m \times n}$ is a vector space
 - Elements of $\mathbb{R}^{m \times n}$ may be added entry-wise
 - Elements can be scaled by elements of $\mathbb{R}$.
 - The zero-matrix acts as an additive identity.
- The canonical basis for $\mathbb{R}^{m \times n}$ is the set of matrices $\{e^{ij}\}$ where

$$(e_{kl}^{ij}) = \delta_{(i,j)(k,l)}$$

Linear functionals

- $\mathbb{R}$ can be viewed as a one-dimensional vector space
- A *linear functional* is a linear map from $\mathbf{V} \rightarrow \mathbb{R}$
- The set of linear functionals is called the *dual space* of $\mathbf{V}$ and is often denoted by $\mathbf{V}^*$
 - $\mathbf{V}^*$ is a vector space under point-wise addition and scalar multiplication.

Linear functionals

- Define $\mathbf{a}_i^* : A \rightarrow \mathbb{R}$ on A by $\mathbf{a}_i^*(\mathbf{a}_j) = \delta_{ij}$ and extend linearly to get a linear functional on $\mathbf{V}$.

$$\mathbf{a}_1^*(\mathbf{a}_1) = 1 \quad \mathbf{a}_2^*(\mathbf{a}_1) = 0 \quad \mathbf{a}_2^*(\mathbf{a}_2) = 1$$

- ▶ $\mathbf{a}_i^*$ returns the i -th coordinate: $\mathbf{a}_i^*(\mathbf{v}) = \alpha_i^v$
- Let $f \in \mathbf{V}^*$ be some linear functional on $\mathbf{V}$, for $\mathbf{v} \in \mathbf{V}$

$$f(\mathbf{v}) = f\left(\sum_{k=1}^n \alpha_k^v \mathbf{a}_k\right) = \sum_{k=1}^n \alpha_k^v f(\mathbf{a}_k) = \sum_{k=1}^n f(\mathbf{a}_k) \mathbf{a}_k^*(\mathbf{v})$$

Linear functionals

- This holds $\forall \mathbf{v} \in \mathbf{V}$. So,

$$f = \sum_{k=1}^n f(\mathbf{a}_k) \mathbf{a}_k^*$$

This is a linear combination of the elements of A^* , so A^* spans $\mathbf{V}^*$. Since it is also linearly independent it is a basis.

- For a basis $A = \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ for $\mathbf{V}$ the corresponding basis $A^* = \{\mathbf{a}_1^*, \dots, \mathbf{a}_n^*\}$ for $\mathbf{V}^*$ is called the *dual basis* of A
- The coordinate representation of f with respect to A^* is then

$$(f(\mathbf{a}_1), \dots, f(\mathbf{a}_n)) \in \mathbb{R}^n$$

Linear functionals

- Define $\psi_A : \mathbf{V}^* \rightarrow \mathbf{V}$ by $\psi_A(f) = \sum_{i=1}^n f(\mathbf{a}_i) \mathbf{a}_i$
 - ψ_A is an isomorphism.
- So any $\mathbf{v}^* \in \mathbb{R}^n$ may be viewed as a linear functional on $\mathbb{R}^n$.
- Write $\mathbf{v}^* = (v_1^*, \dots, v_n^*)$ and $\mathbf{v} = (v_1, \dots, v_n)$. Then

$$\mathbf{v}^*(\mathbf{v}) = \sum_{i=1}^n v_i^* v_i \equiv \langle \mathbf{v}^*, \mathbf{v} \rangle$$

- To evaluate $f(\mathbf{v})$, add up the coordinate-wise products of the coordinate representations of f and $\mathbf{v}$:

$$f(\mathbf{v}) = \sum_{i=1}^n \alpha_i^v f(\mathbf{a}_i)$$

Inner products

- An *inner product* on a real vector space $\mathbf{V}$ is a symmetric, positive definite, bilinear form, i.e., a map $\langle \cdot, \cdot \rangle : \mathbf{V} \times \mathbf{V} \rightarrow \mathbb{R}$ such that
 - *Positive definiteness*, $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ with equality only when $\mathbf{v} = 0$
 - *Symmetry*, $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$
 - *Bilinearity*, $\forall \mathbf{v} \in \mathbf{V}$, the maps $\langle \mathbf{v}, \cdot \rangle : \mathbf{V} \rightarrow \mathbb{R}$ and $\langle \cdot, \mathbf{v} \rangle : \mathbf{V} \rightarrow \mathbb{R}$ are linear. Thus

$$\langle \mathbf{v}, \alpha \mathbf{w}_1 + \beta \mathbf{w}_2 \rangle = \alpha \langle \mathbf{v}, \mathbf{w}_1 \rangle + \beta \langle \mathbf{v}, \mathbf{w}_2 \rangle$$

$$\langle \alpha \mathbf{w}_1 + \beta \mathbf{w}_2, \mathbf{v} \rangle = \alpha \langle \mathbf{w}_1, \mathbf{v} \rangle + \beta \langle \mathbf{w}_2, \mathbf{v} \rangle$$

Inner products and Linear functionals on $\mathbb{R}^n$

For $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, define $\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{i=1}^n v_i w_i$

- If $\mathbf{v} \in \mathbb{R}^n$ then $\mathbf{v}^* = \langle \cdot, \mathbf{v} \rangle \in (\mathbb{R}^n)^*$
- If $\mathbf{v}^* \in (\mathbb{R}^n)^*$ then $\exists \mathbf{v} \in \mathbb{R}^n$ such that $\mathbf{v}^* = \langle \cdot, \mathbf{v} \rangle$
- $\mathbb{R}^n \simeq (\mathbb{R}^n)^*$ with the isomorphism given by $\mathbf{v} \mapsto \langle \cdot, \mathbf{v} \rangle$
 - ψ_A depends on the basis A ; this isomorphism depends only on $\langle \cdot, \cdot \rangle$.
- The kernel of $\mathbf{v}^*$ is the subspace of $\mathbb{R}^n$ orthogonal to $\mathbf{v}$.

Matrix multiplication

- If P is $m \times n$ and Q is $n \times k$ then PQ is $m \times k$, with the ij -entry the inner product of the i -th row of P with the j -th column of Q :

$$(PQ)_{ij} = \sum_{l=1}^n p_{il}q_{lj}$$

- Matrix multiplication is not commutative.
- The reason for this definition is revealed by viewing matrices as linear maps.

Matrices as Linear Maps

- Let $A = \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ and $B = \{\mathbf{b}_1, \dots, \mathbf{b}_m\}$ be bases of $\mathbf{V}$ and $\mathbf{W}$. Let $T : \mathbf{V} \rightarrow \mathbf{W}$ be linear.
- T is determined by its action on A . Each $T(\mathbf{a}_j) \in \mathbf{W}$ has a unique expansion against B :

$$T(\mathbf{a}_j) = \sum_{i=1}^m \beta(T)_{ij} \mathbf{b}_i$$

- ▶ $\beta(T)_j = (\beta(T)_{1j}, \dots, \beta(T)_{mj}) \in \mathbb{R}^m$ is the coordinate representation of $T(\mathbf{a}_j)$ against B .

Matrices as Linear Maps

- Collect the n columns

$$\beta(T) = \begin{pmatrix} \vdots & \vdots & \dots & \vdots \\ \beta(T)_1 & \beta(T)_2 & \dots & \beta(T)_n \\ \vdots & \vdots & \dots & \vdots \end{pmatrix} \in \mathbb{R}^{m \times n}$$

- ▶ Row i indexes B ; column j indexes A .
- The columns of $\beta(T)$ are the coordinate-vectors of the images under T of the basis for $\mathbf{V}$.

Matrices as Linear Maps

- For $\mathbf{v} \in \mathbf{V}$,

$$\begin{aligned} T(\mathbf{v}) &= T\left(\sum_{j=1}^n \alpha_j^v \mathbf{a}_j\right) = \sum_{j=1}^n \alpha_j^v T(\mathbf{a}_j) \\ &= \sum_{j=1}^n \alpha_j^v \left(\sum_{i=1}^m \beta(T)_{ij} \mathbf{b}_i\right) = \sum_{i=1}^m \left(\sum_{j=1}^n \beta(T)_{ij} \alpha_j^v\right) \mathbf{b}_i \end{aligned}$$

- The i -th coordinate of $T(\mathbf{v})$ against B is $\sum_{j=1}^n \beta(T)_{ij} \alpha_j^v$: the i -th row of $\beta(T)$ against α^v .
- So the coordinate representation of $T(\mathbf{v})$ against B is

$$\beta(T)\alpha^v$$

Matrices as Linear Maps

- Matrices are exactly linear maps represented against bases

$$\begin{array}{ccc} \mathbf{V} & \xrightarrow{T} & \mathbf{W} \\ \varphi_A \downarrow & & \uparrow \varphi_B^{-1} \\ \mathbb{R}^n & \xrightarrow{\beta(T)} & \mathbb{R}^m \end{array}$$

- Here φ_A and φ_B are the canonical isomorphisms
- If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $S : \mathbb{R}^m \rightarrow \mathbb{R}^k$ are linear then $\beta(S \circ T) = \beta(S)\beta(T)$: matrix multiplication is composition of linear maps.

Transpose

- The *transpose* M' or $M^{\top}$ of an $m \times n$ matrix M is an $n \times m$ matrix with $m'_{ij} = m_{ji}$
 - Rows and columns are switched.
- If P and Q are conformable then

$$(PQ)' = Q' P'$$

Row vectors are linear functionals

Let $\mathbf{v} \in \mathbb{R}^n$, viewed as a column vector. Then $\mathbf{v}^\top$ or $\mathbf{v}'$ is the corresponding row vector.

$$\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \Rightarrow \mathbf{v}' = (v_1 \ \dots \ v_n)$$

- $\mathbf{v}' \mathbf{w} = \sum_{i=1}^n v_i w_i = \langle \mathbf{v}, \mathbf{w} \rangle$
 - The action of $\mathbf{v}'$ as a linear functional (i.e. matrix multiplication) is as an inner product
- The transpose maps a vector space to its dual space
 - $\mathbf{v} \in \mathbb{R}^n$ implies $\mathbf{v}' \in (\mathbb{R}^n)^*$

Linear maps are columns of linear functionals

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be any function.

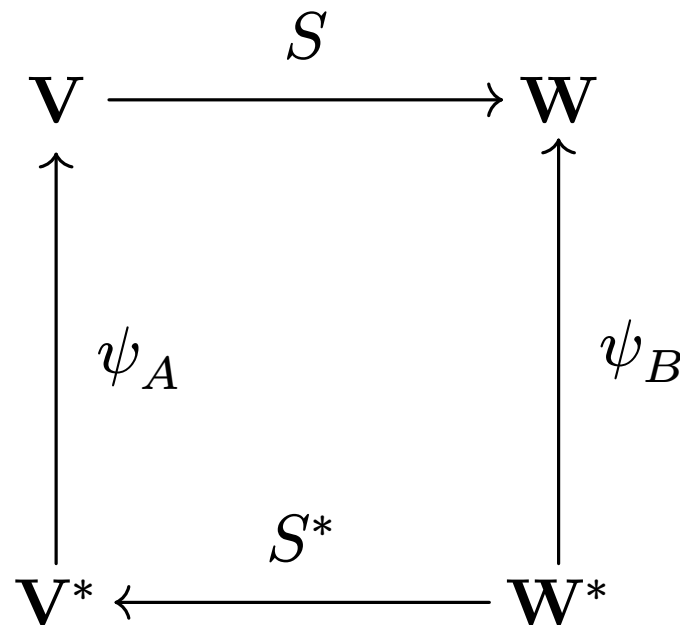
- $f(\mathbf{x}) = \begin{pmatrix} f^1(\mathbf{x}) \\ \vdots \\ f^m(\mathbf{x}) \end{pmatrix}$ where $f^i : \mathbb{R}^n \rightarrow \mathbb{R}$
- If f is linear then each f^i is a linear functional.
- If f is linear then f is a column vector of row vectors.
 - ▶ Row i of $\beta(f)$ is the functional f^i ; column j is $f(\mathbf{e}_j)$.
 - ▶ Transposing turns the functionals into vectors and the images into functionals.

The dual map

- A transpose is a *dual* concept
- If $S : \mathbf{V} \rightarrow \mathbf{W}$ is a linear map between vector spaces then we may define the dual map $S^* : \mathbf{W}^* \rightarrow \mathbf{V}^*$ as,

$\mathbf{w}^* \in \mathbf{W}^*$ implies that $S^*(\mathbf{w}^*) : \mathbf{V} \rightarrow \mathbb{R}$

is given by $S^*(\mathbf{w}^*)(\mathbf{v}) = \mathbf{w}^*(S(\mathbf{v}))$



The dual map

- If bases are chosen for $\mathbf{V}$ and $\mathbf{W}$ and $\beta(S)$ represents S , then against the dual bases

$$\beta(S^*) = \beta(S)'$$

Determinant

- A determinant, $\det$, is a map from $\mathbb{R}^{n \times n} \rightarrow \mathbb{R}$.
- Its properties include
 - $\det(I) = 1$ where I is the identity matrix
 - $\det(PQ) = \det(P) \det(Q)$
 - $\det(\alpha P) = \alpha^n \det(P)$
 - $\det(P) = \det(P')$
- The determinant is the sum of products of entries in the matrix
 - It may be viewed as a polynomial of degree n in its entries
 - Because it is a polynomial in its entries, the determinant is computable

Determinant

- Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be linear and let $S = [0, 1]^n$ be the unit cube.
 - Because T is linear, $T(S)$ is an m -dimensional parallelepiped for some $m \leq n$.
 - $\det(T)$ is the signed volume of this parallelepiped.
- If $m < n$ then the volume of $T(S)$ as a subset of $\mathbb{R}^n$ is zero, and $\det(T) = 0$.
 - By rank-nullity, $m < n$ means $\dim(\ker(T)) \neq 0$.
- **Lemma**
 $\det(T) = 0 \Leftrightarrow \dim(\ker(T)) > 0$

Eigenvalues and eigenspaces

- Let $T : V \rightarrow V$ be linear. λ is an *eigenvalue* of T if there is a $\mathbf{v} \neq 0$ with

$$T(\mathbf{v}) = \lambda \mathbf{v}$$

- Such a $\mathbf{v}$ is an *eigenvector* of T associated to λ .
 - T does not move $\mathbf{v}$ off its own line. It only rescales it.
- Equivalently $(T - \lambda I)(\mathbf{v}) = 0$, so $\mathbf{v} \in \ker(T - \lambda I)$.

Eigenvalues and eigenspaces

- The *eigenspace* associated to λ is

$$\mathbf{E}_\lambda = \ker(T - \lambda I)$$

- ▶ A kernel, hence a subspace: eigenvectors for a given λ , together with 0.
- ▶ λ is an eigenvalue iff $\dim(\mathbf{E}_\lambda) > 0$.
- ▶ $\mathbf{E}_\lambda$ is invariant under T : $T(\mathbf{E}_\lambda) \subseteq \mathbf{E}_\lambda$.
- None of this refers to a basis. Eigenvalues and eigenspaces are properties of the map.

The characteristic polynomial

- By the Lemma, $\dim(\mathbf{E}_\lambda) > 0 \Leftrightarrow \det(T - \lambda I) = 0$, so

$$\chi_T(\lambda) \equiv \det(T - \lambda I) = 0$$

- The determinant is a polynomial of degree n in its entries, so $\chi_T(\lambda)$ is a polynomial of degree n in the indeterminate λ .
- $\chi_T(\lambda)$ is the *characteristic polynomial* of T ; the *eigenvalues* of T are the roots of this polynomial.

The characteristic polynomial

- $\det(T)$ is the product of the eigenvalues of T .
 - The product is real, since non-real eigenvalues come in conjugate pairs.
- So $\det(T) = 0$ iff zero is an eigenvalue of T , iff there is a non-zero $\mathbf{v}$ with $T(\mathbf{v}) = 0$.
- This is the Lemma again: the eigenspace associated to the zero eigenvalue is the kernel.

Invertibility

- Let $f : X \rightarrow Y$ be any map between sets.
 - f is *injective* if $f(x_1) = f(x_2)$ implies $x_1 = x_2$. Injective functions don't lose information.
 - f is *surjective* if for any $y \in Y$ there is an $x \in X$ with $y = f(x)$. The image of X is all of Y .
 - f is *bijective* if both.
- f is bijective if and only if it is invertible.

Invertibility

- A linear map $T : \mathbf{V} \rightarrow \mathbf{W}$ is *invertible* provided there is a linear $T^{-1} : \mathbf{W} \rightarrow \mathbf{V}$ with $T^{-1} \circ T(\mathbf{v}) = \mathbf{v}$.
- Bijectivity is necessary for invertibility; it is also sufficient.
- The following are equivalent:
 - T is invertible
 - $\dim(\mathbf{V}) = \dim(\mathbf{W})$ and $\dim(\ker(T)) = 0$
 - $\dim(\mathbf{V}) = \dim(\mathbf{W}) = \text{rank}(T)$

Invertibility

- A matrix is invertible provided the associated linear map is invertible; the matrix associated to T^{-1} is M^{-1} , with $M^{-1}M = I$.
- The only possible invertible matrices are square with trivial null-space.
- **Theorem**
A square matrix is invertible iff its determinant is non-zero.
 - A square matrix with $\det \neq 0$ is *nonsingular*; otherwise it is *singular*.

Why invertibility matters

- We are often faced with a system of nonlinear equations $f(\mathbf{x}, \mathbf{y}) = 0$ where we need to solve for $\mathbf{y}$; here $\mathbf{x} \in \mathbb{R}^m$ and $\mathbf{y} \in \mathbb{R}^n$.

- Suppose we approximate this with a linear system

$$P\mathbf{y} + Q\mathbf{x} = 0, \quad P \in \mathbb{R}^{n \times n}, \quad Q \in \mathbb{R}^{n \times m}$$

- The determinant is computable. Compute $\det(P)$; if it is non-zero then P^{-1} exists and

$$\mathbf{y} = -P^{-1}Q\mathbf{x}$$

- This is why linear maps are so important: their equations are the only equations we know how to solve.